# Rationality of the center of a generic division algebra with a group action over a field of characteristic 0

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**Abstract**

Let $F$ be a field of characteristic 0 and let $H = C_3 \times C_3$ act regularly as a transitive subgroup of $S_9$. We compute the second integral cohomology group of the restricted Procesi lattice $M|_H$ attached to Problem 5.2 of Auel–Brussel–Garibaldi–Vishne: $\mathrm{H}^2(H, M|_H) \cong C_9$. Its exponent 9 is incompatible both with quasi-permutation and with every finite-index permutation superlattice, while all permutation lattices have $\mathrm{H}^2$-exponent dividing 3; consequently $M|_H$ is neither quasi-permutation nor stably permutation, and its flasque class is not the class of an invertible lattice. This integral cohomological obstruction is of a different nature from the classical unramified Brauer group. *Whether $Z_H(F, 9)$, the center of the generic division algebra of degree 9 with $H$-action, is rational, stably rational, or retract rational is left open*: converting the integral obstruction into a rationality verdict would require the untwisted forms of the Endo–Miyata and flasque-class criteria, which the literature supplies only for twisted invariant fields and which fail in general for the fields $F(M)^G$ (Remark 2.7). The computation generalizes to all odd primes $p$: for $H = C_p \times C_p \subset S_{p^2}$, one obtains $\mathrm{H}^2(H, M|_H) \cong C_{p^2}$ while permutation lattices have exponent dividing $p$; there too no rationality verdict is claimed.

**Keywords:** generic division algebra, stable rationality, integral group cohomology, permutation lattices.

**MSC2020:** 14E08, 20J06, 14M20.

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# 1 Introduction

Let $F$ be a field of characteristic 0. For an integer $n \geq 1$ and a subgroup $H \subseteq S_n$, the *generic division algebra with $H$-action*, denoted $UD(F, n, H)$, has center $Z_H(F, n)$. Problem 5.2 of Auel–Brussel–Garibaldi–Vishne [1, Problem 5.2] asks:

> Determine if $Z_H(F, 9)$ is rational, stably rational, or retract rational over $F$, where $H = C_3 \times C_3 \subset S_9$ is transitive.

Let $G$ be a finite group and let $M$ be a $\mathbb{Z}[G]$-lattice, that is, a finitely generated $\mathbb{Z}$-free $\mathbb{Z}[G]$-module. Write $F(M)$ for the fraction field of the group algebra $F[M]$, that is, the function field of the torus with character lattice $M$, on which $G$ acts through the $\mathbb{Z}[G]$-module structure of $M$; write $F(M)^G$ for its $G$-invariant subfield. For a subgroup $H$ of $G$, write $M|_H$ for $M$ regarded as a $\mathbb{Z}[H]$-lattice by restriction of scalars. Cyclic groups are denoted $C_n$, the augmentation ideal is $I[G] = \ker(\mathbb{Z}[G] \twoheadrightarrow \mathbb{Z})$, where the augmentation map sends $\sum n_g g \mapsto \sum n_g$, and $\exp(A)$ denotes the exponent of a finite abelian group $A$. Write $\mathrm{H}^i(G, M)$ for the $i$-th integral group cohomology of a $G$-module $M$; for $\mathbb{Z}[G]$-lattices this coincides with Tate cohomology in positive degrees [16, Def. 6.2.4].

A field extension $K/F$ is *rational* if $K$ is purely transcendental over $F$, and *stably rational* if $K(t_1, \ldots, t_m)$ is rational over $F$ for some $m \geq 0$.

The rationality problem for fields of the form $F(M)^G$ has a long history. Endo and Miyata [7, §1] developed the theory of quasi-permutation modules and introduced the notion of quasi-rational extensions, providing an algebraic foundation for the rationality problem. Let $G$ be a finite group, $K/k$ a Galois extension with group $G$, and $M$ a $\mathbb{Z}[G]$-lattice. Write $K(M)$ for the fraction field of the group algebra $K[M]$, on which $G$ acts through the $\mathbb{Z}[G]$-module structure of $M$ and the Galois action on $K$. They proved [8, Theorem 1.6] that $K(M)^G$ is *quasi-rational* over $k$ if and only if $M$ is a quasi-permutation $\mathbb{Z}[G]$-lattice. Voskresenskii [15, Chapter 2] gave a comprehensive treatment via the birational geometry of algebraic tori. Colliot-Thélène and Sansuc [5, §1] developed flasque resolutions and used them to characterize stable rationality of such fields in general by the vanishing of the flasque class. Saltman [13, Section 3] introduced retract rationality and constructed the first examples of non-rational stably rational fields.

When $H \subset S_n$ is cyclic and transitive, $Z_H(F, n)$ is rational if $F$ contains a primitive $n$-th root of unity (and retract rational for general $F$ when $n$ is not of the form $8m$) [1, pp. 234]. Thus $H = C_3 \times C_3$ is the first case where the rationality question is genuinely open for $n$ odd.

| *Property* | *Status* | *Method* |
|---|---|---|
| Stably rational | **Open** | $M\|_H$ not quasi-permutation (Lemma 4.7) |
| Rational | **Open** | — |
| Retract rational | **Open** | $\rho(M\|_H)$ not invertible (Proposition 4.11) |

Table 1: Status of the rationality properties of $Z_H(F, 9)$ for $H = C_3 \times C_3$.

In this paper we determine the stable rationality and rationality of $Z_H(F, 9)$:

The main new ingredient is the computation of the second integral cohomology group of the restricted Procesi lattice $M|_H$: $\mathrm{H}^2(H, M|_H) \cong C_9$ (Lemma 4.2). The exponent 9 of this group directly obstructs quasi-permutation: we prove that for any permutation $\mathbb{Z}[H]$-lattice $P$ (Definition 2.1), every element of $\mathrm{H}^2(H, P)$ has order dividing 3 (Proposition 4.4). Consequently $M|_H$ is neither quasi-permutation (Lemma 4.7) nor stably permutation (Theorem 4.6), and the flasque class $\rho(M|_H)$ is not the class of an invertible lattice (Proposition 4.11). This integral cohomological obstruction appears to be new for *multiplicative invariant fields*, i.e. the fields $F(M)^G$ above.

The logical chain is:

$$\begin{aligned} &\mathrm{H}^2(H, M|_H) \cong C_9 \\ &\quad\Longrightarrow\ M|_H \text{ is not quasi-permutation} \qquad \text{(Lemma 4.7)} \\ &\quad\Longrightarrow\ \rho(M|_H) \text{ is not invertible} \qquad \text{(Theorem 4.6, Proposition 4.11).} \end{aligned}$$

Turning the last item into a statement about $Z_H(F, 9)$ would require the untwisted forms of the Endo–Miyata and flasque-class criteria; the literature supplies these criteria only for twisted invariant fields, and they fail in general for the fields $F(M)^G$ (Remark 2.7). *We therefore leave open whether $Z_H(F, 9)$ is rational, stably rational, or retract rational over $F$*: what is proved here is the integral obstruction above and its generalization to all odd primes $p$ (Theorem 5.1).

**Hypotheses.** The original statement in [1, Problem 5.2] is made over an arbitrary field. The cohomological computations ($\mathrm{H}^2(H, M|_H) \cong C_9$ and $\exp(\mathrm{H}^2(H, P)) \mid 3$) are purely integral, valid whenever $\operatorname{char} F \nmid 9$. They yield the lattice-theoretic conclusions of this paper (non-quasi-permutation, non-stable-permutation, and non-invertibility of $\rho(M|_H)$) over any such field. No rationality verdict for $Z_H(F, 9)$ is claimed: Remark 2.7 explains why the two classical criteria are not available in the untwisted setting.

**Notation.** For the main case (Problem 5.2), $H = C_3 \times C_3 = \langle a, b \mid a^3 = b^3 = 1,\ ab = ba \rangle$; the generalization to $H = C_p \times C_p$ is treated in §5. The base field $F$ is of characteristic 0. For a finite group $G$, a $G$-lattice is a finitely generated $\mathbb{Z}[G]$-module that is $\mathbb{Z}$-free. The group ring is denoted $\mathbb{Z}[G]$.

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**Data availability**. The accompanying code package, available at `https://github.com/wuzhengyao/abgv52-lean`, contains the machine-checked Lean development and the two independent verification programs for the cohomological readings used in this paper, written in Python and in GAP using HAP, each with built-in self-tests; it also records their output for $p = 3$ and $p = 5$ and a `SHA-256` manifest.

# 2 Preliminaries

**Definition 2.1.** A $\mathbb{Z}[G]$-*lattice* is a finitely generated $\mathbb{Z}[G]$-module that is $\mathbb{Z}$-free. A $\mathbb{Z}[G]$-lattice $M$ is called:

1. *permutation* if $M \cong \bigoplus_{i=1}^{r} \mathbb{Z}[G/K_i]$ for subgroups $K_i \leq G$, where $\mathbb{Z}[G/K]$ denotes the $\mathbb{Z}[G]$-module with $\mathbb{Z}$-basis the left cosets $G/K$ and $G$-action by left multiplication;

2. *stably permutation* if there exist permutation lattices $P, Q$ such that $M \oplus P \cong Q$;

3. *quasi-permutation* if there is an exact sequence $0 \to M \to P \to Q \to 0$ of $\mathbb{Z}[G]$-lattices in which $P$ and $Q$ are permutation lattices [7, §1];

These concepts are standard; see [5, §1].

**Lemma 2.2.** *Let $H$ be a finite group and $K \trianglelefteq H$ a normal subgroup, and let $\mathbb{Z}[H/K]$ be the permutation $\mathbb{Z}[H]$-lattice (Definition 2.1). The character of the rational representation $\mathbb{Q}[H/K]$ is*

$$\chi_{H/K}(g) = \big|\{xK \in H/K : gxK = xK\}\big| = \begin{cases} [H:K], & g \in K, \\ 0, & g \notin K. \end{cases}$$

This lemma is used in the proof of Lemma 3.8 below, where the character of the permutation module $\mathbb{Q}[H]$ is evaluated with $K = \{1\}$.

The next definitions are those of Colliot-Thélène and Sansuc [5, §1].

**Definition 2.3.** Let $G$ be a finite group and let $M$ be a $\mathbb{Z}[G]$-lattice. Then $M$ is *flasque* if $\hat{\mathrm{H}}^{-1}(K, M) = 0$ for every subgroup $K \leq G$, and *coflasque* if $\mathrm{H}^1(K, M) = 0$ for every subgroup $K \leq G$.

**Definition 2.4.** Two $\mathbb{Z}[G]$-lattices $M$ and $N$ are *similar* if $M \oplus S_1 \cong N \oplus S_2$ for some permutation lattices $S_1$ and $S_2$. This is an equivalence relation; the similarity class of $M$ is written $[M]$. A $\mathbb{Z}[G]$-lattice is *invertible* if it is a direct summand of a permutation lattice [5, §1].

**Proposition 2.5.** *Every $\mathbb{Z}[G]$-lattice $M$ admits a* flasque resolution

$$0 \longrightarrow M \longrightarrow P \longrightarrow F \longrightarrow 0$$

*with $P$ permutation and $F$ flasque [5, §1, Lemme 3]. The similarity class $[F]$ depends only on $M$ [5, §1, Lemme 5]; it is written $\rho(M)$ and called the* flasque class *of $M$.*

**Lemma 2.6.** *Let $G$ be a finite group. Let $M$ be a $\mathbb{Z}[G]$-lattice which is a direct summand of a permutation lattice. Then $M$ is coflasque.*

*Proof.* Let $M \oplus M' \cong P$ with $P$ permutation. Cohomology is additive, so $\mathrm{H}^1(G, M)$ is a direct summand of $\mathrm{H}^1(G, P)$. It therefore suffices to show $\mathrm{H}^1(G, P) = 0$ for $P$ permutation.

Write $P \cong \bigoplus_i \mathbb{Z}[G/K_i]$ with $K_i \leq G$. By Shapiro's Lemma [16, Theorem 6.3.2], $\mathbb{Z}[G/K_i] \cong \mathrm{Ind}_{K_i}^G(\mathbb{Z})$ and hence $\mathrm{H}^1(G, \mathbb{Z}[G/K_i]) \cong \mathrm{H}^1(K_i, \mathbb{Z})$. Since $K_i$ acts trivially on $\mathbb{Z}$, $\mathrm{H}^1(K_i, \mathbb{Z}) \cong \mathrm{Hom}(K_i, \mathbb{Z}) = 0$. Therefore $\mathrm{H}^1(G, P) = 0$. □

**Remark 2.7.** Two criteria are used in the literature for the *twisted* invariant field $Q(L/K, M) = L(M)^G$, where $L/K$ is Galois with group $G$ and $G$ acts on the coefficient field as well: *stable rationality implies quasi-permutation* [8, Theorem 1.6] [14, Theorem 1.3(b)], and *retract rationality if and only if $\rho(M)$ is the class of an invertible lattice* [13, Theorem 3.14(a)]. For the untwisted field $F(M)^G = F(M, G)$ of [14, p. 221] neither statement is available, and both fail in general.

(*i*) *Stable rationality does not force quasi-permutation.* Let $G = C_2 \times C_2$ and $M = M_2(G)$, which is a faithful $G$-lattice [14]; then $F(M, G)$ is the center of the generic $G$-crossed product, and by a theorem of Snider this center is rational over $\mathbb{Q}$. On the other hand $\eta(M_2(G)) \neq [0]$ [13, Theorems 2.6 and 2.7], so $\rho(M)$ is not the class of an invertible lattice and $M$ is not a quasi-permutation lattice. The same phenomenon is recorded in [14, p. 226]: "still assuming $F(G)/F$ is retract rational, $F(M, G)/F$ can be retract rational for more $M$."

(*ii*) *Consequently the integral obstruction of this paper does not yield a rationality verdict for $Z_H(F, 9)$.* What it yields is that $M|_H$ is neither quasi-permutation nor stably permutation and that $\rho(M|_H)$ is not invertible (Lemma 4.7, Theorem 4.6, Proposition 4.11).

In the language of quasi-permutation lattices the twisted criterion is [8, Theorem 1.6]; see also [7, §1].

In the sequel we prove that $M|_H$ is *not* quasi-permutation: the cohomological computation of §4 gives $\mathrm{H}^2(H, M|_H) \cong C_9$ while permutation lattices have $\mathrm{H}^2$-exponent dividing 3, so $M|_H$ is neither quasi-permutation (Lemma 4.7) nor stably permutation (Theorem 4.6). By Remark 2.7 this does not by itself decide the rationality type of $Z_H(F, 9)$. The same computation generalizes to $C_p \times C_p$ in §5.

# 3 The Procesi Lattice

We give a self-contained construction of the Procesi $\mathbb{Z}[S_9]$-lattice $M$ and the Procesi isomorphism, keeping only what is needed for the rationality computations. An accessible modern exposition is in [1, §5] and [9]; the original reference is [12].

## 3.1 Construction over $S_9$

Let $n = 9$ and let $\Omega = S_9/S_8$ be the set of left cosets of $S_8$ in $S_9$. Then $|\Omega| = 9$ and $S_9$ acts on $\Omega$ by left multiplication. Define the permutation

$\mathbb{Z}[S_9]$-module

$$U = \mathbb{Z}[\Omega] \cong \bigoplus_{\omega\in\Omega} \mathbb{Z}\cdot e_\omega,$$

where $\{e_\omega\}_{\omega\in\Omega}$ is the standard basis and $g\cdot e_\omega = e_{g\omega}$ for $g\in S_9$. Consider the $\mathbb{Z}[S_9]$-module map

$$\varphi : U\otimes_{\mathbb{Z}} U \longrightarrow \mathbb{Z}[\Omega], \qquad \varphi(e_\alpha\otimes e_\beta) = e_\alpha - e_\beta,$$

extended $\mathbb{Z}$-linearly. The map $\varphi$ is $S_9$-equivariant:

$$\varphi\big(g\cdot(e_\alpha\otimes e_\beta)\big) = \varphi(e_{g\alpha}\otimes e_{g\beta}) = e_{g\alpha}-e_{g\beta} = g\cdot(e_\alpha - e_\beta) = g\cdot\varphi(e_\alpha\otimes e_\beta).$$

The image of $\varphi$ is the *augmentation submodule*

$$I[\Omega] = \ker\Big(\varepsilon : \mathbb{Z}[\Omega]\twoheadrightarrow\mathbb{Z},\ \varepsilon\big(\sum a_\omega e_\omega\big) = \sum a_\omega\Big).$$

Indeed, every difference $e_\alpha - e_\beta$ has augmentation zero, and conversely any element $\sum a_\omega e_\omega$ with $\sum a_\omega = 0$ is a $\mathbb{Z}$-linear combination of such differences (take $\{e_\alpha - e_{\omega_0} : \alpha\neq\omega_0\}$ for a fixed $\omega_0$). Moreover, $\varphi$ is surjective onto $I[\Omega]$.

**Definition 3.1.** The *Procesi* $\mathbb{Z}[S_9]$*-lattice* is $M = \ker(\varphi)$.

By construction we have the short exact sequence of $\mathbb{Z}[S_9]$-modules

$$0\longrightarrow M\longrightarrow U\otimes_{\mathbb{Z}} U \xrightarrow{\ \varphi\ } I[\Omega]\longrightarrow 0. \tag{1}$$

The module $M$ is $\mathbb{Z}$-free: it is a $\mathbb{Z}$-submodule of the free $\mathbb{Z}$-module $U\otimes_{\mathbb{Z}}U$, and every submodule of a free $\mathbb{Z}$-module is free (since $\mathbb{Z}$ is a principal ideal domain) [4, Ch.VII, §3, Th.1, Cor.2]. The ranks are

$$\operatorname{rk}_{\mathbb{Z}} U = 9, \qquad \operatorname{rk}_{\mathbb{Z}}(U\otimes U) = 81, \qquad \operatorname{rk}_{\mathbb{Z}} I[\Omega] = 8, \qquad \operatorname{rk}_{\mathbb{Z}} M = 73.$$

**Proposition 3.2.** *[12]; see also [1, §5] and [9]. Let $F$ be a field of characteristic 0, with $M$ and $U$ as above, and let $Z_{S_9}(F,9)$ be the center of the generic matrix algebra of degree 9 over $F$.*

1. *The center satisfies $Z_{S_9}(F,9)\cong F(M\oplus U)^{S_9}$, where $F(M\oplus U)^{S_9}$ is the invariant field of the purely transcendental extension $F(M\oplus U)$ under the natural $S_9$-action induced by the $\mathbb{Z}[S_9]$-module structure of $M\oplus U$.*

2. *For every subgroup $H\subseteq S_9$, taking $H$-invariants gives $Z_H(F,9) = F((M\oplus U)|_H)^H$.*

3. *If $H$ acts faithfully on $M|_H$, then $Z_H(F,9)$ has transcendence degree $82 = n^2+1$ over $F$ and is* rational *over $F(M|_H)^H$ [10, Proposition 1.4].*

The $S_9$ identification of this proposition, and the rationality statement of its third part for $H = C_3 \times C_3$, are completely formalized in Lean in the accompanying code package; the structure the statement speaks about is constructed there, so the statement is not vacuous, and the identification requires no input beyond characteristic 0.

## 3.2 Restriction to $H = C_3 \times C_3$

Now fix a transitive embedding $H = C_3 \times C_3 \hookrightarrow S_9$, i.e., an embedding whose image acts transitively on $\Omega = S_9/S_8$. Since $|H| = 9 = |\Omega|$, the action is moreover regular (i.e., free and transitive, called *simply transitive* in [3, ch.I, §5, no.6, def.7]): choosing a basepoint $\omega_0 \in \Omega$, the map $H \to \Omega$, $h \mapsto h \cdot \omega_0$, is an $H$-equivariant bijection. Consequently we obtain an explicit isomorphism of $\mathbb{Z}[H]$-modules

$$U|_H \xrightarrow{\cong} \mathbb{Z}[H], \qquad e_{h\cdot\omega_0} \longmapsto h \quad (h \in H). \tag{2}$$

**Proposition 3.3.** *The isomorphism* (2) *is independent of the choice of* $\omega_0$ *up to a* $\mathbb{Z}[H]$*-module automorphism of* $\mathbb{Z}[H]$*. Consequently,* $\mathrm{Ext}^1_{\mathbb{Z}[H]}(-, M|_H)$*,* $\mathrm{H}^2(H, M|_H)$*, and the isomorphism class of* $M|_H$ *are independent of this choice.*

*Proof.* Let $\omega_0' = g \cdot \omega_0$ be another basepoint. The isomorphism induced by $\omega_0'$ sends $e_{h\cdot\omega_0'} = e_{hg\cdot\omega_0} \longmapsto h$, while the original sends $e_{hg\cdot\omega_0} \longmapsto hg$; hence the two isomorphisms differ by right multiplication by $g^{-1}$ on $\mathbb{Z}[H]$. Right multiplication by a fixed element of $H$ is a $\mathbb{Z}[H]$-module automorphism of the left regular module. Therefore the induced map on cohomology and $\mathrm{Ext}^1$ is the identity, and the isomorphism class of $M|_H$ is unchanged. □

**Proposition 3.4.** *Let* $\iota_1, \iota_2 : H \hookrightarrow S_9$ *be two transitive embeddings. Then* $M|_{\iota_1(H)} \cong M|_{\iota_2(H)}$ *as* $\mathbb{Z}[H]$*-modules.*

*Proof.* Both $\iota_1$ and $\iota_2$ make $\Omega$ into a regular $H$-set (since $|H| = 9 = |\Omega|$ and each action is transitive). Any two regular $H$-sets of the same cardinality are $H$-equivariantly isomorphic: fix basepoints $\omega_1, \omega_2 \in \Omega$, and define a bijection $\Omega \to \Omega$ by $h \cdot \omega_1 \longmapsto h \cdot \omega_2$. This bijection is an element $\sigma \in S_9$, and by construction $\sigma\, \iota_1(h)\, \sigma^{-1} = \iota_2(h)$ for all $h \in H$. Conjugating the embedding by $\sigma$ replaces the $H$-action on $U \otimes U$ with its conjugate under the $S_9$-action, which induces an isomorphism $M|_{\iota_1(H)} \cong M|_{\iota_2(H)}$ of $\mathbb{Z}[H]$-modules via the identification of $H$ with $\iota_1(H)$ and $\iota_2(H)$ followed by the conjugation $\iota_2^{-1} \circ \iota_1$. □

**Proposition 3.5.** *There is an isomorphism of $\mathbb{Z}[H]$-modules*

$$\mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H] \;\cong\; \mathbb{Z}[H]^{\oplus 9},$$

*where the left side carries the diagonal action $h\cdot(x\otimes y) = hx\otimes hy$ and the right side carries the action $h\cdot(z_b)_{b\in H} = (hz_{h^{-1}b'})_{b'\in H}$, that is, left multiplication composed with the coordinate permutation $b\mapsto hb$.*

*Proof.* We use the group basis $\{h : h\in H\}$ of $\mathbb{Z}[H]$, so that $\mathbb{Z}[H]\otimes_{\mathbb{Z}}\mathbb{Z}[H]$ carries the diagonal action $h\cdot(x\otimes y) = hx\otimes hy$; we write $y = \sum_{a\in H} y_a\, a$ with $y_a\in\mathbb{Z}$. Define

$$\Phi : \mathbb{Z}[H]\otimes_{\mathbb{Z}}\mathbb{Z}[H] \longrightarrow \mathbb{Z}[H]^{\oplus 9}, \qquad \Phi(x\otimes y) = (\, y_a\, x\,)_{a\in H},$$

so that the $a$-th coordinate of $\Phi(x\otimes y)$ is $y_a x$. On a basis vector $f\otimes g$ this reads $\Phi(f\otimes g) = f$ in the $g$-th coordinate; hence

$$\Psi\big((z_a)_{a\in H}\big) \;=\; \sum_{a\in H} z_a\otimes a,$$

viewed as a $\mathbb{Z}$-linear map $\Psi : \mathbb{Z}[H]^{\oplus 9}\to\mathbb{Z}[H]\otimes_{\mathbb{Z}}\mathbb{Z}[H]$, is inverse to $\Phi$: both composites are the identity on basis vectors (on the tensor side $\Phi(z_b\otimes b) = z_b$ in the $b$-th coordinate, and $\Psi(\Phi(x\otimes y)) = \sum_a x\otimes(y_a a) = x\otimes y$). Finally $(hy)_a = y_{h^{-1}a}$ for all $a\in H$, so

$$\Phi\big(h\cdot(x\otimes y)\big) = \Phi(hx\otimes hy) = (\, y_{h^{-1}a}\, hx\,)_{a\in H} = h\cdot(\, y_a\, x\,)_{a\in H} = h\cdot\Phi(x\otimes y)$$

for the action on the right side, and $\Phi$ is an isomorphism of $\mathbb{Z}[H]$-modules. □

Under the identification (2) and Proposition 3.5, the restriction of the map $\varphi$ to $H$ yields the following.

**Proposition 3.6.** *There is an exact sequence of $\mathbb{Z}[H]$-modules*

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus 9} \xrightarrow{\pi|_H} I[H] \longrightarrow 0, \tag{3}$$

*where $I[H] = \ker(\varepsilon : \mathbb{Z}[H]\twoheadrightarrow\mathbb{Z})$, $\varepsilon(\sum a_h h) = \sum a_h$, is the augmentation ideal of $H$. All modules in the sequence are finitely generated and $\mathbb{Z}$-free, and* $\mathrm{rk}_{\mathbb{Z}}\, M|_H = 73$.

*Proof. Exactness.* Under the identifications, the map $\varphi$ becomes a $\mathbb{Z}[H]$-module map $\pi|_H : \mathbb{Z}[H]^{\oplus 9}\longrightarrow I[H]$. Each standard basis element of $\mathbb{Z}[H]^{\oplus 9}$ maps to a difference $h-h'$ in $\mathbb{Z}[H]$; these differences generate the $\mathbb{Z}$-module $I[H]$, and $I[H]$ is generated as a $\mathbb{Z}[H]$-module by any single element $h-1$

with $h \neq 1$; hence $\pi|_H$ is surjective. Define $M|_H = \ker(\pi|_H)$, the restriction of the Procesi lattice to $H$.

*Freeness.* All modules in the sequence are finitely generated and $\mathbb{Z}$-free: $\mathbb{Z}[H]^{\oplus 9}$ is $\mathbb{Z}$-free with basis the group elements in each coordinate; $I[H]$ is a $\mathbb{Z}$-submodule of $\mathbb{Z}[H]$, hence free [4, Ch.VII, §3, Th.1, Cor.2]; $M|_H = \ker(\pi|_H)$ is a $\mathbb{Z}$-submodule of $\mathbb{Z}[H]^{\oplus 9}$, hence free.

*Rank.* $\operatorname{rk}_{\mathbb{Z}} M|_H = \operatorname{rk}_{\mathbb{Z}} \mathbb{Z}[H]^{\oplus 9} - \operatorname{rk}_{\mathbb{Z}} I[H] = 9 \cdot 9 - 8 = 73$, matching the rank of $M$ over $S_9$. □

**Proposition 3.7.**

$$M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \;\cong\; 8 \cdot \mathbb{Q}[H] \;\oplus\; \mathbb{Q},$$

*where $\mathbb{Q}$ denotes the one-dimensional trivial $\mathbb{Q}[H]$-module.*

*Proof.* Tensor the exact sequence of Proposition 3.6 with $\mathbb{Q}$ (flat over $\mathbb{Z}$, [2, Ch.II, §2, no.4, Th.1]):

$$0 \longrightarrow M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathbb{Q}[H]^{\oplus 9} \longrightarrow I[H] \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow 0.$$

The augmentation sequence

$$0 \longrightarrow I[H] \longrightarrow \mathbb{Z}[H] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0 \tag{4}$$

remains exact after tensoring with $\mathbb{Q}$ (flat over $\mathbb{Z}$, [2, Ch.II, §2, no.4, Th.1]), giving

$$0 \longrightarrow I[H] \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathbb{Q}[H] \xrightarrow{\varepsilon} \mathbb{Q} \longrightarrow 0.$$

This sequence splits since $|H| = 9$ is invertible in $\mathbb{Q}$ (Maschke's Theorem [6, (3.14)]; the splitting is given by the idempotent $e_0 = \frac{1}{|H|} \sum_{h \in H} h$).

We now compute $\ker \pi|_H$ explicitly. Write $z_b = \sum_{x \in H} (z_b)_x \, x$ with $(z_b)_x \in \mathbb{Q}$ for an element $z$ of $\mathbb{Q}[H]^{\oplus 9}$, and define

$$\Theta(z)_t \;=\; \sum_{x \in H} (z_{tx})_x \, x, \qquad t \in H.$$

The coefficient identity $(\Theta(z)_t)_y = (z_{ty})_y$ is equivalent to $(z_b)_y = ((\Theta z)_{by^{-1}})_y$; hence $\Theta^{-1}(W)_b = \sum_{x \in H} (W_{bx^{-1}})_x \, x$ is inverse to $\Theta$, with

$$(\Theta^{-1}(\Theta(z))_b)_y = (\Theta(z)_{by^{-1}})_y = (z_b)_y, \quad (\Theta(\Theta^{-1}(W))_t)_y = (\Theta^{-1}(W)_{ty})_y = (W_t)_y.$$

Moreover

$$\big(\Theta(h \cdot z)_t\big)_y = (z_{h^{-1}ty})_{h^{-1}y} = \big(h \cdot \Theta(z)_t\big)_y,$$

using commutativity of $H$ for $h^{-1}ty = t\,h^{-1}y$; hence $\Theta$ is an isomorphism from $\mathbb{Q}[H]^{\oplus 9}$ with the action $h\cdot(z_b) = (h\,z_{h^{-1}b})$ to $\mathbb{Q}[H]^{\oplus 9}$ with the coordinatewise action $h\cdot(z_b) = (h\,z_b)$. Under this identification the map $\pi|_H$ corresponds to

$$\pi_{\mathrm{std}}(z) \;=\; \sum_{b\in H} z_b\,(1-b)\,:$$

indeed $\Theta^{-1}$ fixes the standard coordinate vectors $u_b$ (the vector with the unit 1 in coordinate $b$), and $\pi|_H(u_b) = 1-b$, since under (2) and Proposition 3.5 the vector $u_b$ corresponds to $e_{\omega_0}\otimes e_{b\cdot\omega_0}$, whose image under $\varphi$ is the difference $e_{\omega_0} - e_{b\cdot\omega_0}$. Both sides are $\mathbb{Q}[H]$-linear for the coordinatewise action (for $\pi_{\mathrm{std}}$ this follows since $\pi|_H$ is $\mathbb{Z}[H]$-linear and $\Theta$ is an isomorphism for the two structures), and they agree on the $\mathbb{Q}[H]$-basis $\{u_b\}_{b\in H}$; hence they coincide. Consequently, identifying $M|_H\otimes_{\mathbb{Z}}\mathbb{Q}$ with the kernel of the $\mathbb{Q}$-linear extension of $\pi|_H$ to $\mathbb{Q}[H]^{\oplus 9}$, the map $\Theta$ restricts to an isomorphism $M|_H\otimes_{\mathbb{Z}}\mathbb{Q}\;\cong\; E = \ker\pi_{\mathrm{std}}$, a $\mathbb{Q}[H]$-submodule of $\mathbb{Q}[H]^{\oplus 9}$ for the coordinatewise action. It remains to decompose $E$.

Let

$$\pi_1\colon \mathbb{Q}[H]^{\oplus 9}\longrightarrow \mathbb{Q}[H]^{\oplus 8}\times\mathbb{Q},\qquad z\longmapsto \big((z_b)_{b\neq 1},\;\varepsilon(z_1)\big),$$

$$\sigma\colon \mathbb{Q}[H]^{\oplus 8}\longrightarrow \mathbb{Q}[H],\qquad y\longmapsto \sum_{b\neq 1} y_b\,(1-b),$$

which are $\mathbb{Q}[H]$-linear for the coordinatewise action ($\varepsilon(hz_1)=\varepsilon(z_1)$). Put $K=\ker\sigma$ and $\widetilde{I} = \{z : z_b = 0 \text{ for } b\neq 1,\; z_1\in I[H]\otimes\mathbb{Q}\}$. Since $\pi_{\mathrm{std}}(z) = \sigma((z_b)_{b\neq 1})$, we have $E=\pi_1^{-1}(K\times\mathbb{Q})$, and $\pi_1$ restricts to an exact sequence of $\mathbb{Q}[H]$-modules

$$0\longrightarrow \widetilde{I}\longrightarrow E\xrightarrow{\pi_1} K\times\mathbb{Q}\longrightarrow 0.$$

Indeed $\pi_1$ maps $E$ onto $K\times\mathbb{Q}$: for $y\in K$ and $c\in\mathbb{Q}$ the element with coordinates $y$ on $b\neq 1$ and $c\cdot 1$ on $b=1$ lies in $E$ and is mapped to $(y,c)$. Its kernel on $E$ consists of the $z$ with $z_b=0$ for $b\neq 1$ and $\varepsilon(z_1)=0$, that is, of $\widetilde{I}$. Moreover the elements $1-b$, $b\neq 1$, form a $\mathbb{Q}$-basis of $I[H]\otimes\mathbb{Q}$ (they span, as $I[H]$ is generated by differences, and are linearly independent: the coefficient of $y\neq 1$ in $\sum_{b\neq 1} c_b\,(1-b)$ is $-c_y$); hence $\sigma$ is surjective and

$$0\longrightarrow K\longrightarrow \mathbb{Q}[H]^{\oplus 8}\xrightarrow{\sigma} I[H]\otimes\mathbb{Q}\longrightarrow 0$$

is exact. Both exact sequences split, because $\mathbb{Q}[H]$ is semisimple by Maschke [6, (3.14)] and all modules involved are finitely generated. Finally $\widetilde{I}\;\cong$

$I[H]\otimes\mathbb{Q}$ as $\mathbb{Q}[H]$-modules via $z \mapsto z_1$ (with inverse the coordinate inclusion). Combining the splittings,

$$E \;\cong\; \widetilde{I} \oplus K \oplus \mathbb{Q} \;\cong\; (I[H] \otimes \mathbb{Q}) \oplus K \oplus \mathbb{Q} \;\cong\; \mathbb{Q}[H]^{\oplus 8} \oplus \mathbb{Q},$$

and, recalling that $M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \cong E$ under $\Theta$, this proves that $M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \cong 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$ as $\mathbb{Q}[H]$-modules. □

From Proposition 3.7 it follows that if $M|_H$ were a permutation $\mathbb{Z}[H]$-lattice, then $M|_H \cong 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$ (Lemma 3.8 below).

**Lemma 3.8.** *Let $P$ be a permutation $\mathbb{Z}[H]$-lattice satisfying $P \otimes_{\mathbb{Z}} \mathbb{Q} \cong 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$. Then $P \cong 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$.*

*Proof.* A permutation $\mathbb{Z}[H]$-lattice is, equivalently, the permutation module $\mathbb{Z}[X]$ of a finite $H$-set $X$ (Definition 2.1). By flatness of $\mathbb{Q}$ over $\mathbb{Z}$ and Proposition 3.7,

$$P \otimes_{\mathbb{Z}} \mathbb{Q} \;\cong\; 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$$

as $\mathbb{Q}[H]$-modules. For $g \in H$, the trace of $g$ on $8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$ equals $8 \cdot \#\{h \in H : gh = h\} + 1$, where $\#\{h : gh = h\}$ is 9 for $g = 1$ and 0 otherwise (Lemma 2.2 with $K = \{1\}$); this is 73 for $g = 1$ and 1 for $g \neq 1$. On the other hand, the trace of $g$ on $P \otimes \mathbb{Q}$ equals the trace of $g$ on the permutation module $\mathbb{Z}[X]$, which is the number of fixed points of $g$ on $X$. Comparing, $|X| = 73$ and $g$ has exactly one fixed point in $X$ for every $g \neq 1$.

Let $g \neq 1$ and let $x_g$ be its unique fixed point in $X$. For every $h \in H$ the point $hx_g$ is again fixed by $g$, since $g(hx_g) = h(gx_g) = hx_g$ by commutativity of $H$; uniqueness forces $hx_g = x_g$. Thus $x_g$ is fixed by all of $H$. Fix $g_0 \neq 1$, which exists as $|H| = 9$. Its fixed point $x_{g_0}$ is then the unique $H$-fixed point of $X$: an $H$-fixed point is fixed by $g_0$, hence equals $x_{g_0}$. Write $x_0 = x_{g_0}$.

If $x \in X \setminus \{x_0\}$ were fixed by some $g \neq 1$, then $x = x_g = x_0$, a contradiction. Hence $H$ acts freely on $X \setminus \{x_0\}$, so this set is a disjoint union of orbits of cardinality $|H| = 9$; as $|X \setminus \{x_0\}| = 72 = 8 \cdot 9$, it is the disjoint union of eight free orbits. Therefore $X \cong \{x_0\} \sqcup H^{\sqcup 8}$ as $H$-sets, and $P \cong \mathbb{Z}[X] \cong \mathbb{Z} \oplus 8 \cdot \mathbb{Z}[H]$. □

# 4 Cohomological Obstruction to Quasi-Permutation

In this section we compute the second integral cohomology group $\mathrm{H}^2(H, M|_H)$ and prove that its exponent obstructs quasi-permutation.

**Notation.** Throughout this section we write $\mathrm{H}^q(G, M)$ for the $q$-th integral group cohomology of a $G$-module $M$. The upright H is reserved for the cohomology functor, while the italic $H$ denotes the group $H = C_3 \times C_3$.

## 4.1 Cohomology Computations

**Lemma 4.1.** *The group* $\mathrm{H}^1(H, I[H])$ *is cyclic of order* 9.

*Proof.* Apply group cohomology $\mathrm{H}^*(H, -)$ to the augmentation exact sequence (4). The associated long exact cohomology sequence reads

$$\begin{aligned} 0 \longrightarrow \mathrm{H}^0(H, I[H]) \longrightarrow \mathrm{H}^0(H, \mathbb{Z}[H]) \longrightarrow \mathrm{H}^0(H, \mathbb{Z}) \\ \longrightarrow \mathrm{H}^1(H, I[H]) \longrightarrow \mathrm{H}^1(H, \mathbb{Z}[H]) \longrightarrow \cdots . \end{aligned}$$

We analyse the low-degree terms:

- $\mathrm{H}^0(H, \mathbb{Z}[H]) = \mathbb{Z}[H]^H = \mathbb{Z} \cdot N_H$, where $N_H = \sum_{h \in H} h$ is the norm element.
- $\mathrm{H}^0(H, \mathbb{Z}) = \mathbb{Z}^H = \mathbb{Z}$.
- The map $\mathrm{H}^0(H, \mathbb{Z}[H]) \to \mathrm{H}^0(H, \mathbb{Z})$ sends $N_H \mapsto \varepsilon(N_H) = \sum_{h \in H} 1 = |H| = 9$.
- $\mathrm{H}^1(H, \mathbb{Z}[H]) = 0$: $\mathbb{Z}[H] \cong \mathrm{Ind}_1^H \mathbb{Z}$ as $\mathbb{Z}[H]$-modules, and Shapiro's Lemma [16, Theorem 6.3.2] gives $\mathrm{H}^1(H, \mathrm{Ind}_1^H \mathbb{Z}) \cong \mathrm{H}^1(\{1\}, \mathbb{Z}) = 0$.

From the exactness at $\mathrm{H}^0(H, \mathbb{Z})$ we obtain $\operatorname{coker}(\mathbb{Z} \cdot N_H \xrightarrow{\varepsilon} \mathbb{Z}) = \mathbb{Z}/9\mathbb{Z}$, and exactness at $\mathrm{H}^1(H, I[H])$ with the vanishing of $\mathrm{H}^1(H, \mathbb{Z}[H])$ yields the short exact sequence

$$0 \longrightarrow \mathbb{Z}/9\mathbb{Z} \longrightarrow \mathrm{H}^1(H, I[H]) \longrightarrow 0,$$

hence $\mathrm{H}^1(H, I[H]) \cong \mathbb{Z}/9\mathbb{Z} \cong C_9$. □

**Lemma 4.2.** *For the Procesi restriction* $M|_H$, $\mathrm{H}^2(H, M|_H) \cong C_9$.

*Proof.* Apply $\mathrm{H}^*(H, -)$ to the exact sequence of Proposition 3.6:

$$0 \longrightarrow M|_H \longrightarrow \mathbb{Z}[H]^{\oplus 9} \longrightarrow I[H] \longrightarrow 0.$$

The associated long exact cohomology sequence gives

$$\begin{aligned} \cdots \longrightarrow \mathrm{H}^1(H, \mathbb{Z}[H]^{\oplus 9}) \longrightarrow \mathrm{H}^1(H, I[H]) \\ \longrightarrow \mathrm{H}^2(H, M|_H) \longrightarrow \mathrm{H}^2(H, \mathbb{Z}[H]^{\oplus 9}) \longrightarrow \cdots . \end{aligned}$$

By Shapiro's Lemma [16, Theorem 6.3.2], $\mathrm{H}^i(H, \mathbb{Z}[H]) \cong \mathrm{H}^i(\{1\}, \mathbb{Z}) = 0$ for all $i > 0$. Hence $\mathrm{H}^1(H, \mathbb{Z}[H]^{\oplus 9}) = \mathrm{H}^2(H, \mathbb{Z}[H]^{\oplus 9}) = 0$, and the long exact sequence collapses to the isomorphism

$$\mathrm{H}^2(H, M|_H) \;\cong\; \mathrm{H}^1(H, I[H]).$$

By Lemma 4.1, $\mathrm{H}^1(H, I[H]) \cong C_9$. □

**Lemma 4.3.** *With the trivial $H$-action on $\mathbb{Z}$, $\mathrm{H}^2(H,\mathbb{Z}) \cong C_3 \times C_3$.*

*Proof.* Apply $\mathrm{H}^*(H,-)$ to the short exact sequence of trivial $H$-modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0.$$

The associated long exact sequence reads

$$\begin{aligned} 0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} & \\ \longrightarrow \mathrm{H}^1(H,\mathbb{Z}) \longrightarrow \mathrm{H}^1(H,\mathbb{Q}) \longrightarrow \mathrm{H}^1(H,\mathbb{Q}/\mathbb{Z}) & \\ \longrightarrow \mathrm{H}^2(H,\mathbb{Z}) \longrightarrow \mathrm{H}^2(H,\mathbb{Q}) \longrightarrow \cdots . & \end{aligned}$$

Since $\mathbb{Q}$ is uniquely divisible, multiplication by 3 is an automorphism of $\mathbb{Q}$; hence $\mathrm{H}^i(H,\mathbb{Q}) = 0$ for all $i > 0$ [11, Proposition 1.6.2(c)]. Substituting $\mathrm{H}^1(H,\mathbb{Q}) = \mathrm{H}^2(H,\mathbb{Q}) = 0$ yields $\mathrm{H}^2(H,\mathbb{Z}) \cong \mathrm{H}^1(H,\mathbb{Q}/\mathbb{Z})$. Since $\mathbb{Q}/\mathbb{Z}$ carries the trivial $H$-action, $\mathrm{H}^1(H,\mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(H,\mathbb{Q}/\mathbb{Z})$ [16, Corollary 6.4.6]. Now $\mathrm{Hom}(H,\mathbb{Q}/\mathbb{Z}) \cong C_3 \times C_3$ because $\mathrm{Hom}(C_3,\mathbb{Q}/\mathbb{Z}) \cong C_3$ ($\mathbb{Q}/\mathbb{Z}$ has a unique subgroup of order 3) and $\mathrm{Hom}(-,\mathbb{Q}/\mathbb{Z})$ preserves finite direct sums [3, Ch.II, §1, no.6, Prop.6, Cor.1]. Hence $\mathrm{H}^2(H,\mathbb{Z}) \cong C_3 \times C_3$. □

## 4.2 Exponent of $\mathrm{H}^2$ for Permutation Lattices

**Proposition 4.4.** *Let $P$ be a permutation $\mathbb{Z}[H]$-lattice. Then every element of $\mathrm{H}^2(H,P)$ has order dividing 3; equivalently, $\exp(\mathrm{H}^2(H,P)) \mid 3$.*

*Proof.* Write $P \cong \bigoplus_i \mathbb{Z}[H/K_i]$ for subgroups $K_i \leq H$. Since $\mathrm{H}^2(H,-)$ is additive and therefore preserves finite direct sums [16, Proposition 3.3.4], $\mathrm{H}^2(H,P) \cong \bigoplus_i \mathrm{H}^2(H,\mathbb{Z}[H/K_i])$. By Shapiro's Lemma [16, Theorem 6.3.2], $\mathbb{Z}[H/K_i] \cong \mathrm{Ind}_{K_i}^H(\mathbb{Z})$, hence $\mathrm{H}^2(H,\mathbb{Z}[H/K_i]) \cong \mathrm{H}^2(K_i,\mathbb{Z})$, where $\mathbb{Z}$ carries the trivial $K_i$-action. For each subgroup $K \leq H$ we compute $\mathrm{H}^2(K,\mathbb{Z})$:

- $K = \{1\}$: $\mathrm{H}^2(\{1\},\mathbb{Z}) = 0$.
- $K \cong C_3$ (four subgroups of order 3): $\mathrm{H}^2(C_3,\mathbb{Z}) \cong \mathrm{Hom}(C_3,\mathbb{Q}/\mathbb{Z}) \cong C_3$.
- $K = H = C_3 \times C_3$: $\mathrm{H}^2(H,\mathbb{Z}) \cong C_3 \times C_3$ (Lemma 4.3).

In every case $\exp(\mathrm{H}^2(K,\mathbb{Z})) \mid 3$. Hence $\exp(\mathrm{H}^2(H,P)) \mid 3$. □

**Proposition 4.5.** *If a $\mathbb{Z}[H]$-lattice $L$ is stably permutation, then*

$$\exp(\mathrm{H}^2(H,L)) \mid 3.$$

*Proof.* Write $L \oplus P \cong Q$ with $P, Q$ permutation. By additivity [16, Proposition 3.3.4], $\mathrm{H}^2(H,L) \oplus \mathrm{H}^2(H,P) \cong \mathrm{H}^2(H,Q)$. Both $\mathrm{H}^2(H,P)$ and $\mathrm{H}^2(H,Q)$ have exponent dividing 3 by Proposition 4.4. For any $x \in \mathrm{H}^2(H,L)$, the element $(x,0) \in \mathrm{H}^2(H,Q)$ satisfies $3(x,0) = 0$; hence $3x = 0$. Therefore $\exp(\mathrm{H}^2(H,L)) \mid 3$. □

## 4.3 The Obstruction

**Theorem 4.6.** *The restricted Procesi lattice $M|_H$ is not a stably permutation $\mathbb{Z}[H]$-lattice.*

*Proof.* By Lemma 4.2, $\mathrm{H}^2(H, M|_H) \cong C_9$, a group of exponent 9. If $M|_H$ were stably permutation, Proposition 4.5 would force $\exp(\mathrm{H}^2(H, M|_H)) \mid 3$. Contradiction. □

**Lemma 4.7.** *The lattice $M|_H$ is not quasi-permutation (Definition 2.1).*

*Proof.* Suppose there are permutation lattices $P$, $Q$ and a short exact sequence $0 \to M|_H \to P \to Q \to 0$. The associated long exact cohomology sequence gives

$$\mathrm{H}^1(H, Q) \longrightarrow \mathrm{H}^2(H, M|_H) \longrightarrow \mathrm{H}^2(H, P).$$

The group $\mathrm{H}^1(H, Q)$ vanishes: by Shapiro's lemma [16, Theorem 6.3.2], $\mathrm{H}^1(H, \mathbb{Z}[H/K]) \cong \mathrm{H}^1(K, \mathbb{Z}) = 0$ for every subgroup $K \leq H$. Hence $\mathrm{H}^2(H, M|_H)$ embeds into $\mathrm{H}^2(H, P)$, a group of exponent dividing 3 (Proposition 4.4). This is impossible since $\mathrm{H}^2(H, M|_H) \cong C_9$ has exponent 9 (Lemma 4.2). □

**Remark 4.8.** *Whether $Z_H(F, 9)$ is rational, stably rational, or retract rational over $F$ is left open.* What the preceding results establish is the lattice-theoretic statement: $M|_H$ is not quasi-permutation (Lemma 4.7), not stably permutation (Theorem 4.6), and $\rho(M|_H)$ is not the class of an invertible lattice (Proposition 4.11). By Remark 2.7 none of these converts into a rationality verdict for $F(M|_H)^H$, and $Z_H(F, 9)$ is rational over $F(M|_H)^H$ (Proposition 3.2), the action of $H$ on $M|_H$ being faithful (Proposition 4.9).

**Proposition 4.9.** *Let $M_9(F)$ denote the algebra of $9 \times 9$ matrices over $F$. Let $H = C_3 \times C_3$ act on $M_9(F)$ via the regular representation $H \hookrightarrow S_9$ by permutation of matrix indices: $h \cdot e_{ij} = e_{h(i),h(j)}$ for the standard matrix units. The action of $H$ on $M|_H$ is the restriction of the diagonal left-regular action on $\mathbb{Z}[H]^{\oplus 9}$ of Proposition 3.5:*

$$h \cdot (z_b)_{b \in H} = (h\, z_{h^{-1}b'})_{b' \in H}, \qquad h \in H,\ (z_b)_{b \in H} \in M|_H \subset \mathbb{Z}[H]^{\oplus 9},$$

*that is, left multiplication composed with the coordinate permutation $b \mapsto hb$. This action is faithful.*

*Proof.* We show that the action on $M|_H$ is faithful. Suppose that $h \in H$ acts trivially on $M|_H$. Since $M|_H = \ker(\pi|_H)$ by Proposition 3.6, the difference map

$$\delta_h(z) \;=\; h \cdot x - x, \qquad x \in \mathbb{Z}[H]^{\oplus 9} \text{ any lift of } z \in I[H],$$

is well defined on $I[H] = \mathrm{im}(\pi|_H)$: if $x - x' \in \ker(\pi|_H) = M|_H$, then $h \cdot (x - x') = x - x'$. Identify $\mathbb{Z}[H]^{\oplus 9} \cong \mathbb{Z}[H] \otimes_{\mathbb{Z}} \mathbb{Z}[H]$ with basis $\{\delta_b\}_{b \in H}$ in the second factor, as in Proposition 3.5; the diagonal action reads $h \cdot (a \otimes \delta_b) = ha \otimes \delta_{hb}$, and, as in the Second proof below,

$$\pi|_H\Big(\sum_{b \in H} z_b \otimes \delta_b\Big) = \sum_{b \in H} \big(z_b - \varepsilon(z_b)\, b\big).$$

Choose $g \in H \setminus \{1, h, h^{-1}\}$, which exists because $|H| = 9 > 3$. We compute $\delta_h(g-1)$ in two ways. First, $g \otimes \delta_1$ maps to $g - 1$ under $\pi|_H$, so

$$\delta_h(g-1) = h \cdot (g \otimes \delta_1) - g \otimes \delta_1 = hg \otimes \delta_h - g \otimes \delta_1.$$

Second, $-(1 \otimes \delta_g)$ also maps to $-\big(1 - \varepsilon(1)\, g\big) = g - 1$, so

$$\delta_h(g-1) = -h \cdot (1 \otimes \delta_g) + 1 \otimes \delta_g = 1 \otimes \delta_g - h \otimes \delta_{hg}.$$

The $\delta_{hg}$-component of the right-hand side of the first display is 0, because $\delta_{hg} \neq \delta_h$ and $\delta_{hg} \neq \delta_1$ (as $g \neq 1$ and $g \neq h^{-1}$), whereas in the second display it is $-h \neq 0$. This contradiction shows that no $h \neq 1$ acts trivially on $M|_H$; the action is faithful. □

*Second proof.* Let $h \in H$, $h \neq 1$, and let $z \in \mathbb{Z}[H]^{\oplus 9}$ be the element with coordinates $z_1 = h - 1$, $z_h = 1 - h$, and $z_b = 0$ for $b \notin \{1, h\}$. Under the identifications of Proposition 3.5, the element with coordinates $(z_b)_{b \in H}$ corresponds to $\sum_{b \in H} z_b \otimes \delta_b$, where $\delta_b$ denotes the group element $b$ of the second factor, identified with $e_{b \cdot \omega_0}$. Since $\varphi(e_\alpha \otimes e_\beta) = e_\alpha - e_\beta$, the map $\pi|_H$ sends this element to $\sum_{b \in H}(z_b - \varepsilon(z_b)\, b)$. Since $\varepsilon(h-1) = \varepsilon(1-h) = 0$, this gives $\pi|_H(z) = (h-1) + (1-h) = 0$, and $z \in M|_H$. Now act by $h$. By Proposition 3.5 the $h^{-1}$-th coordinate of $z$ moves into the first coordinate and is multiplied by $h$, so $(h \cdot z)_1 = h \cdot z_{h^{-1}}$. Here $h^{-1} \neq 1$ because $h \neq 1$, and $h^{-1} \neq h$: if $h^{-1} = h$, then $h^2 = 1$, and $h = h^3(h^2)^{-1} = 1$ since $h^3 = 1$. Hence $z_{h^{-1}} = 0$, and $(h \cdot z)_1 = 0$, whereas $z_1 = h - 1 \neq 0$. Therefore $h \cdot z \neq z$, that is, $h$ acts nontrivially on $M|_H$. □

**Remark 4.10.** For the *twisted* invariant field $Q(L/K, M) = L(M)^G$ the flasque-class criterion is a theorem: $Q(L/K, M)$ is retract rational over $K$ if and only if $\rho(M)$ is the class of an invertible lattice [13, Theorem 3.14(a)]. For the untwisted field $F(M)^G = F(M, G)$ of [14, p. 221] the corresponding statement is not available and fails in general, by the example of Remark 2.7: for $G = C_2 \times C_2$ and $M = M_2(G)$ the field $F(M, G)$ is rational, whereas $\rho(M)$ is not the class of an invertible lattice. We therefore do not use this criterion; from it we keep only the hypothesis-free half proved below, namely that $\rho(M|_H)$ is not the class of an invertible lattice (Proposition 4.11).

**Proposition 4.11.** *Let $H = C_3 \times C_3$. Then the flasque class $\rho(M|_H)$ is not the class of an invertible lattice; more precisely, $\rho(M|_H)$ is not coflasque.*

*Proof.* We show that $\rho(M|_H)$ is not coflasque; then Lemma 2.6 gives the claim.

Let $P$ be a permutation $\mathbb{Z}[H]$-lattice containing $M|_H$ with finite index. Then $P$ and $M|_H$ have the same rank, so Proposition 3.7 gives

$$P \otimes_{\mathbb{Z}} \mathbb{Q} \;\cong\; M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \;\cong\; 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}.$$

Write $P \cong \bigoplus_{K \le H} m_K \cdot \mathbb{Z}[H/K]$ with $m_K \ge 0$. The multiplicity of the regular representation in $P \otimes_{\mathbb{Z}} \mathbb{Q}$ equals $m_{\{1\}}$, hence $m_{\{1\}} = 8$. The multiplicity of the trivial representation equals $\sum_K m_K$, hence $\sum_K m_K = 9$. The rank of $P$ equals $\sum_K m_K[H:K]$, hence

$$73 \;=\; \sum_K m_K[H:K] \;=\; 72 \;+\; \sum_{K \neq \{1\}} m_K[H:K].$$

Since $\sum_{K \neq \{1\}} m_K = 1$, the remaining orbit has index 1, that is, $K = H$. Therefore

$$P \;\cong\; \mathbb{Z}[H]^{\oplus 8} \oplus \mathbb{Z}.$$

By Shapiro's Lemma and Lemma 4.3,

$$\mathrm{H}^2(H, P) \;\cong\; \mathrm{H}^2(H, \mathbb{Z}) \;\cong\; C_3 \times C_3,$$

and in particular $\exp(\mathrm{H}^2(H,P)) \mid 3$ for every such $P$. On the other hand $\mathrm{H}^2(H, M|_H) \cong C_9$ by Theorem 4.6.

Let $E = P/M|_H$. The short exact sequence $0 \to M|_H \to P \to E \to 0$ induces an exact sequence

$$\mathrm{H}^1(H, E) \longrightarrow \mathrm{H}^2(H, M|_H) \xrightarrow{\alpha} \mathrm{H}^2(H, P).$$

The group $\mathrm{H}^2(H, M|_H)$ is cyclic of order 9, and $\mathrm{H}^2(H, P)$ has exponent dividing 3. Hence the image of $\alpha$ has order dividing 3, so $\ker(\alpha) \neq 0$. By exactness $\mathrm{H}^1(H, E) \neq 0$, and $E$ is not coflasque.

Now let $0 \to M|_H \to P \to F \to 0$ be a flasque resolution of $M|_H$ (Proposition 2.5). Its middle term $P$ is a permutation lattice containing $M|_H$ with finite index, so the preceding paragraph applies to it, and $F$ is not coflasque. Since an invertible lattice is coflasque (Lemma 2.6), the class $\rho(M|_H) = [F]$ is not the class of an invertible lattice. By Remark 4.10 this carries no rationality verdict for the untwisted field $F(M|_H)^H$. □

**Proposition 4.12.** *The restricted Procesi lattice $M|_H$ is not a permutation $\mathbb{Z}[H]$-lattice.*

*Proof.* Apply $\mathrm{Hom}_{\mathbb{Z}[H]}(-, L)$ to (4). The associated long exact sequence reads

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}, L) &\longrightarrow \mathrm{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}[H], L) \longrightarrow \mathrm{Hom}_{\mathbb{Z}[H]}(I[H], L) \\ &\longrightarrow \mathrm{Ext}^1_{\mathbb{Z}[H]}(\mathbb{Z}, L) \longrightarrow \mathrm{Ext}^1_{\mathbb{Z}[H]}(\mathbb{Z}[H], L) \longrightarrow \mathrm{Ext}^1_{\mathbb{Z}[H]}(I[H], L) \\ &\longrightarrow \mathrm{Ext}^2_{\mathbb{Z}[H]}(\mathbb{Z}, L) \longrightarrow \mathrm{Ext}^2_{\mathbb{Z}[H]}(\mathbb{Z}[H], L) \longrightarrow \cdots . \end{aligned}$$

Since $\mathbb{Z}[H]$ is projective, $\mathrm{Ext}^i_{\mathbb{Z}[H]}(\mathbb{Z}[H], -) = 0$ for all $i \geq 1$. Substituting this collapses the sequence to an isomorphism $\mathrm{Ext}^1_{\mathbb{Z}[H]}(I[H], L) \cong \mathrm{Ext}^2_{\mathbb{Z}[H]}(\mathbb{Z}, L) = \mathrm{H}^2(H, L)$. Taking $L = M|_H$ and $L = 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$ yields

$$\mathrm{Ext}^1_{\mathbb{Z}[H]}(I[H], M|_H) \cong C_9, \qquad \mathrm{Ext}^1_{\mathbb{Z}[H]}(I[H], 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}) \cong C_3 \times C_3.$$

By Proposition 3.7, $M|_H \otimes_{\mathbb{Z}} \mathbb{Q} \cong 8 \cdot \mathbb{Q}[H] \oplus \mathbb{Q}$. If $M|_H$ were a permutation lattice, Lemma 3.8 would then give $M|_H \cong 8 \cdot \mathbb{Z}[H] \oplus \mathbb{Z}$, and the two groups above would coincide, contradicting $C_9 \ncong C_3 \times C_3$. □

*Second proof.* By Lemma 4.2, $\mathrm{H}^2(H, M|_H) \cong C_9$, a group of exponent 9. If $M|_H$ were a permutation $\mathbb{Z}[H]$-lattice, Proposition 4.4 would force $\exp(\mathrm{H}^2(H, M|_H)) \mid 3$. Contradiction. □

**Remark 4.13.** Proposition 4.12 is strictly weaker than Theorem 4.6, and the latter is strictly weaker than Lemma 4.7. Indeed a permutation lattice is stably permutation (take $P = 0$) and a stably permutation lattice is quasi-permutation (take the sequence $0 \to M \to Q \to P \to 0$ associated with $M \oplus P \cong Q$), and both inclusions are strict: for $G = C_2 \times C_2$ the direct sum $\chi_1 \oplus \chi_2$ of two distinct non-trivial characters is stably permutation since $\chi_1 \oplus \chi_2 \oplus \mathbb{Z}[G/\ker(\chi_1\chi_2)] \cong \mathbb{Z}[G]$, while it is not permutation: the character of every non-zero permutation lattice $\mathbb{Z}[G/K]$ contains the trivial character, whereas $\chi_1 + \chi_2$ does not; and for $G = C_3$ the augmentation ideal $I[C_3]$ is quasi-permutation via $0 \to I[C_3] \to \mathbb{Z}[C_3] \to \mathbb{Z} \to 0$, but not stably permutation: passing to the coinvariants $- \otimes_{\mathbb{Z}[C_3]} \mathbb{Z}$ maps it onto $C_3$, while permutation lattices over the cyclic group $C_3$ have torsion-free coinvariants. Thus non-permutation does not imply non-stable permutation, and non-stable permutation does not imply non-quasi-permutation. The main results of this paper therefore use Lemma 4.7 and Theorem 4.6 alone; the implication "stably rational ⇒ quasi-permutation" for the untwisted field is not available (Remark 2.7).

# 5 Generalization to $C_p \times C_p$

We record the generalization of the lattice-theoretic results above to all odd primes.

**Theorem 5.1.** *Let $p$ be an odd prime and let $H = C_p \times C_p \subset S_{p^2}$ act regularly (i.e., freely and transitively on $\{1, 2, \ldots, p^2\}$). Let $M|_H$ be the restriction of the Procesi $\mathbb{Z}[S_{p^2}]$-lattice to $H$. Let $F$ be a field of characteristic 0, and let $Z_H(F, p^2)$ denote the center of the generic division algebra of degree $p^2$ over $F$ (cf. Proposition 3.2), rational over $F(M|_H)^H$. Then:*

1. $\mathrm{H}^2(H, M|_H) \cong C_{p^2}$.
2. *For any permutation $\mathbb{Z}[H]$-lattice $P$,* $\exp(\mathrm{H}^2(H, P)) \mid p$.
3. $M|_H$ *is not quasi-permutation.*
4. $M|_H$ *is not stably permutation.*
5. *The rationality type of $Z_H(F, p^2)$ (rational, stably rational, or retract rational) is left open; no verdict is claimed (Remark 2.7).*

*Proof.* The Procesi restriction to $H$ follows the same construction as in §3: the regular identification $U|_H \cong \mathbb{Z}[H]$ (Proposition 3.3) and the isomorphism $\mathbb{Z}[H] \otimes \mathbb{Z}[H] \cong \mathbb{Z}[H]^{\oplus p^2}$ (Proposition 3.5, whose proof works verbatim with 9 replaced by $p^2$) hold integrally for all $p$. The map $\pi|_H : \mathbb{Z}[H]^{\oplus p^2} \to I[H]$ is surjective because $I[H]$ is generated by $h - 1$ ($h \neq 1$), and $M|_H = \ker(\pi|_H)$ is $\mathbb{Z}$-free (submodule of a free $\mathbb{Z}$-module) of $\mathbb{Z}$-rank $p^4 - p^2 + 1$.

(1) The same cohomological computation as in Lemmas 4.1 and 4.2, with 9 replaced by $p^2$ and $H = C_p \times C_p$, gives $\mathrm{H}^2(H, M|_H) \cong C_{p^2}$.

(2) The same argument as in Proposition 4.4, with the subgroups of $C_p \times C_p$ in place of those of $C_3 \times C_3$, gives $\exp(\mathrm{H}^2(H, P)) \mid p$.

(3) The same argument as in Lemma 4.7, with $p$ in place of 3, shows that $M|_H$ is not quasi-permutation.

(4) The same argument as in Theorem 4.6, with $p$ in place of 3, shows that $M|_H$ is not stably permutation.

(5) No rationality verdict is claimed, by Remark 2.7. □

**Remark 5.2.** The proof of Theorem 5.1 holds for all odd primes $p$ without any dependence on $p = 3$ specifically: the regular embedding $H \hookrightarrow S_{p^2}$ (independent of the choice up to conjugation; Proposition 3.4), Proposition 3.5, the cohomological vanishing (Shapiro), and the subgroup analysis of $C_p \times C_p$ are all purely formal and independent of $p$. As explained in Remark 2.7, the corresponding untwisted criteria are unavailable, which is why no rationality verdict for $Z_H(F, p^2)$ is claimed.

Our results on [1, Problem 5.2] are summarized in Table 1.

# References


[1] A. Auel, E. Brussel, S. Garibaldi, and U. Vishne. Open problems on central simple algebras. *Transform. Groups*, 16(1):219–264, 2011. 3, 4, 7, 8, 20

[2] Nicolas Bourbaki. *Éléments de mathématique. Algèbre commutative. Chapitres 1 à 4.* Berlin: Springer, reprint of the 1985 original edition, 2006. 11

[3] Nicolas Bourbaki. *Éléments de mathématique. Algèbre. Chapitres 1 à 3.* Berlin: Springer, reprint of the 1970 original edition, 2007. 9, 15

[4] Nicolas Bourbaki. *Éléments de mathématique. Algèbre. Chapitres 4 à 7.* Berlin: Springer, reprint of the 1981 original edition, 2007. 8, 11

[5] J.-L. Colliot-Thélène and J.-J. Sansuc. La R-équivalence sur les tores. *Ann. Sci. Éc. Norm. Supér. (4)*, 10:175–229, 1977. 3, 6

[6] C. W. Curtis and I. Reiner. Methods of representation theory, with applications to finite groups and orders. Vol. I. Pure and Applied Mathematics. A Wiley-Interscience Publication. New York etc.: John Wiley & Sons. XXI, 819 p. £ 40.70 (1981)., 1981. 11, 12

[7] S. Endo and T. Miyata. Quasi-permutation modules over finite groups. *J. Math. Soc. Japan*, 25:397–421, 1973. 3, 6, 7

[8] Shizuo Endo and Takehiko Miyata. Invariants of finite Abelian groups. *J. Math. Soc. Japan*, 25:7–26, 1973. 3, 7

[9] E. Formanek. The ring of generic matrices. *J. Algebra*, 258(1):310–320, 2002. 7, 8

[10] Jr. Lenstra, H. W. Rational functions invariant under a finite abelian group. *Invent. Math.*, 25:299–325, 1974. 8

[11] Jürgen Neukirch, Alexander Schmidt, and Kay Wingberg. *Cohomology of number fields*, volume 323 of *Grundlehren Math. Wiss.* Berlin: Springer, 2nd ed. edition, 2020. Corrected version 2.3, May 2020, electronic edition. 15

[12] C. Procesi. Non-commutative affine rings. *Atti Accad. Naz. Lincei, Mem., Cl. Sci. Fis. Mat. Nat., VIII. Ser., Sez. I*, 8:239–255, 1967. 7, 8

[13] D. J. Saltman. Retract rational fields and cyclic Galois extensions. *Isr. J. Math.*, 47:165–215, 1984. 3, 7, 17

[14] D. J. Saltman. Multiplicative field invariants. *J. Algebra*, 106:221–238, 1987. 7, 17

[15] V. E. Voskresenskiĭ. *Algebraic groups and their birational invariants. Transl. from the original Russsian manuscript by Boris Kunyavskii.*, volume 179 of *Transl. Math. Monogr.* Providence, RI: American Mathematical Society, rev. version of 'Algebraic tori', Nauka 1977 edition, 1998. 3

[16] C. A. Weibel. *An introduction to homological algebra*, volume 38 of *Camb. Stud. Adv. Math.* Cambridge: Cambridge University Press, 1994. 3, 6, 14, 15, 16